

# IMEN266 Operations Research II

## Chapters 1-3. Basic Probability

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INDUSTRIAL AND MANAGEMENT  
ENGINEERING, POSTECH

# Outline

- 1 Basic probability
- 2 Basic reliability
- 3 Bayes' rule
- 4 Random variable

## How to study this chapter (2026 format)

- **Slides** (this file): concepts + class-problem statements. No transcription needed — everything typed lives in the Companion.
- **Notebooks** — every class problem, twice: in class the INCLASS version (fill the TODOs, `check()` answers instantly); at home the full version with solutions, sliders, experiments.
- **Companion** (gap-fill  $\rightarrow$  full): all definitions and theorems + 6 core proofs with blanks, checkpoints, AI study cards.
- **Homework**: A exam-style (no AI first) · B critique an AI solution · C stress-test an assumption · D prompt log.
- Class rhythm (75 min): quiz 5'  $\rightarrow$  concepts  $\rightarrow$  live notebook  $\rightarrow$  wrap-up.
- Ground truth = analytic  $\leftrightarrow$  simulation agreement — not AI confidence.
- Materials: PLMS and GitHub `imen266-2026`.

# What is it?

- Measure of uncertainty
- $\Omega$ : sample space
- Sample point: an outcome of an experiment
- Event: a subset of  $\Omega$
- Example) Roll a dice

# Properties of events and probability measure

- ❶  $P(\Omega) = 1, P(\emptyset) = 0$
- ❷ If  $E_1, E_2, \dots, E_n$  are mutually exclusive (that is  $E_i \cap E_j = \emptyset$  if  $i \neq j$ ),

$$P\left(\bigcup_{i=1}^n E_i\right) = \sum_{i=1}^n P(E_i).$$

$\Rightarrow$  If  $E_1, \dots$  are mutually exclusive,

$$P\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} P(E_i).$$

- ❸ Let  $A$  be an event and  $A^c$  its complement (not in  $A$ ). Then,

$$P(A^c) = 1 - P(A).$$

# Properties of events and probability measure

- ④ Let  $A$  and  $B$  be two events,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

- ⑤ Inclusion-exclusion law

If  $E_1, \dots, E_n$  are events,

$$\begin{aligned} P\left(\bigcup_{i=1}^n E_i\right) &= \sum_{i=1}^n P(E_i) - \sum_{i < j} P(E_i \cap E_j) \\ &\quad + \sum_{i < j < k} P(E_i \cap E_j \cap E_k) \\ &\quad + \cdots + (-1)^{n+1} P(E_1 \cap \cdots \cap E_n). \end{aligned}$$

▷ Two-event proof + general idea: Companion, Proof 1.

# Conditional probability

- Let  $A$  and  $B$  be two events.  $P(A|B)$  is the probability of  $A$  given  $B$ .

$$P(A|B) = \frac{P(A \cap B)}{P(B)}.$$

- Let  $A$ ,  $B$  and  $C$  be three events.

▷ Multiplication rule, typed: Companion §1.2.

# Independence of events

- Events  $A$  and  $B$  are independent iff

$$P(A \cap B) = P(A)P(B).$$

$$\Rightarrow P(A|B) = P(A)$$

- Events  $E_1, \dots, E_n$  are (mutually) independent if

$$P(E_1 \cap \dots \cap E_n) = P(E_1) \dots P(E_n).$$

- Events  $E_1, \dots, E_n$  are pairwise independent if

$$P(E_i \cap E_j) = P(E_i)P(E_j) \text{ for } i \neq j.$$

- Mutually independent  $\Leftrightarrow$  pairwise independent?



# Independence of events

- Example: Toss a coin twice.

▷ Mutual vs pairwise, typed: Companion §1.2.

# Series system

- Consider a system of  $n$  components in series.
- Components are all independent.
- $p_i$ : probability that component  $i$  functions
- What is the probability that the system functions?

► Colab notebook: CP1 warm-up (series/parallel)

# Parallel system

- Same as series system except the system is composed of  $n$  parallel components.
- What is the probability that the system functions?

► Colab notebook: CP1 warm-up (series/parallel)

# Law of total probability

- Let  $B_1, \dots, B_n$  be  $n$  *mutually exclusive* and *exhaustive* events.
  - exhaustive:
  - mutually exclusive:
- Consider an event  $A$  in  $\Omega$  such that
  - $P(A|B_1), \dots, P(A|B_n)$  are known, and
  - $P(B_1), \dots, P(B_n)$  are known.
- Law of total probability

$$P(A) = P(A|B_1)P(B_1) + \dots + P(A|B_n)P(B_n).$$

▷ Statement + proof: Companion, Proof 2.

# Law of total probability

- Four machines produce a total of 10 parts.

|                    | M1   | M2   | M3  | M4  |
|--------------------|------|------|-----|-----|
| # of parts         | 2    | 4    | 1   | 3   |
| prob. of defective | 0.15 | 0.05 | 0.2 | 0.1 |

- One part from 10 is randomly selected. What is the probability that it is defective?

# Class problem 1

1. Three cooks - A, B, and C - bake a special kind of cake, and with respective probabilities 0.02, 0.03, and 0.05 it fails to rise. In the restaurant where they work A bakes 50% of these cakes, B bakes 30%, and C bakes 20%. What fraction of the cakes fail to rise in the restaurant? What is the probability that a given failed cake was baked by cook A?

► Colab notebook: CP1 §1 (cooks)

# Bayes' rule

- $B_1, \dots, B_n$  are mutually exclusive and exhaustive given  $P(B_1), \dots, P(B_n), P(A|B_1), \dots, P(A|B_n)$ .
- Bayes' rule

$$P(B_i|A) = \frac{P(A|B_i)P(B_i)}{\sum_{j=1}^n P(A|B_j)P(B_j)}.$$

▷ Derivation: Companion, Proof 2 — posterior  $\propto$  prior  $\times$  likelihood.

# Bayes' rule

- Example: COVID TEST

There is a test which is positive 99% of the time if a person has virus. The test is positive 2% of the time if a person does not have virus. Assume that 1% of population has virus. For a person, if the test is positive, what is the probability that they have virus?



# Class problem 1

2. Suppose that there was a cancer diagnostic test that was 95% accurate on those that do have cancer and 96% accurate on those who don't. If 0.4% of the population have cancer, compute the probability that a tested person has cancer, given that his or her test results indicate so.

► Colab notebook: CP1 §2 (cancer test)

# Class problem 1

3. Define events appropriately and use Bayes' rule to solve this problem. A gambler has in his pocket a fair coin and a two-headed coin. (a) He selects one of the coins at random, and when he flips it, it shows heads. What is the probability that it is the fair coin? (b) Suppose that he flips the same coin a second time and again it shows heads. Now what is the probability that it is the fair coin? (c) Suppose that he flips the same coin a third time and it shows tails. Now what is the probability that this is the fair coin?

► Colab notebook: CP1 §3 (two coins)

# Random variable

- A function that maps  $\Omega$  to  $\mathbb{R}$ .

# Discrete random variable

- A discrete random variable  $X$  is characterized by its probability mass function (pmf)  $p(x)$ .

$$p(x) = P(X = x).$$

- Example: the probability of producing a defective item is 0.05. Suppose 4 items are produced. Let  $X$  denote the number of defectives.

▷ All named distributions, one table: Companion §1.6 (zoo).

# Class problem 1

4. A company has 7 machines on its shop-floor of which 4 are lathes. The service personnel pick two machines at random and check to see if they have any maintenance problems. Let  $X$  be the number of lathes selected. Find the probability mass function (or probability distribution) of  $X$ . Verify that all properties of PMFs are satisfied.

► Colab notebook: CP1 §4 (lathes)

# Example distributions

## 1. Bernoulli random variable

$X$  is a Bernoulli random variable if it takes value 1 with probability  $p$  and 0 with probability  $1 - p$ .

$$p(x) = \begin{cases} p & \text{if } x = 1 \\ 1 - p & \text{if } x = 0. \end{cases}$$

## 2. Binomial random variable

Consider an experiment comprising  $n$  Bernoulli trials. Let  $X$  denote the number of 1's in  $n$  trials.

$$p(x) = P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}.$$

# Example distributions

## 3. Geometric random variable

Consider an experiment where Bernoulli trials are performed until a success (or 1) is obtained. Let  $X$  denote the number of trials until a success.

$$p(x) = P(X = x) = (1 - p)^{x-1}p.$$

## 4. Negative binomial random variable

Bernoulli trials are performed until  $k$  successes (or 1) are obtained. Let  $X$  denote the number of trials until  $k$  successes.

$$p(x) = P(X = x) = \binom{x-1}{k-1} (1-p)^{x-k} p^k.$$

# Example distributions

## 5. Poisson random variable

$X$  is a Poisson random variable with pmf  $p(x)$  if

$$p(x) = P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}, \text{ for } x \in \mathbb{Z} + .$$



# Class problem 1

5. Historical data has revealed that a manufacturer's current yield is 85%. That is 15% of the parts they produce are defective. If 50 parts are produced.
- (a) What is the probability that exactly 8 parts are defective in total?
  - (b) What is the probability that the 10th part produced is the first defective item?
  - (c) What is the probability that the 20th part produced is the third defective part?

► Colab notebook: CP1 §5 (yield: binom/geom/negbin)

# Class problem 1

6. Cars arrive at a certain yield sign on a road according to a Poisson distribution with  $\lambda = 8$  cars in a minute.
- (a) What is the probability that exactly 5 cars arrive in a minute?
  - (b) What is the probability that more than 2 cars arrive in a minute?

► Colab notebook: CP1 §6 (Poisson cars)

# Continuous random variable

- A continuous random variable  $X$  is such that

$$P(X = x) = 0 \text{ for any } x \in \mathbb{R}.$$

- Cumulative distribution function (cdf)

$$F(x) = P(X \leq x).$$

- Probability density function (pdf)

$$f(x) = \frac{d}{dx}F(x).$$

- Properties

▷ cdf properties, typed: Companion §1.5.

# Class problem 2

1. Compute  $F(x)$ , the cdf if  $X$  is a continuous r.v. with pdf:

$$f(x) = \begin{cases} \frac{3}{4}(1 - x^2) & \text{if } -1 < x < 1, \\ 0 & \text{otherwise.} \end{cases}$$

Also obtain the probability  $P(-0.5 < X < 0.75)$ .

► Colab notebook: CP2 §1 (polynomial pdf)

# Example distributions

1. Exponential random variable: a random variable  $X$  is exponentially distributed with  $\lambda$  if its cdf and pdf are given by

$$F(x) = P(X \leq x) = \begin{cases} 1 - e^{-\lambda x} & \text{if } x > 0 \\ 0 & \text{otherwise.} \end{cases}$$

$$f(x) = \frac{d}{dx}F(x) = \begin{cases} \lambda e^{-\lambda x} & \text{if } x > 0 \\ 0 & \text{otherwise.} \end{cases}$$

2. Erlang random variable: a random variable  $X$  is according to an Erlang distribution with  $\lambda$  and  $k$  if its cdf and pdf are

$$F(x) = P(X \leq x) = \begin{cases} 1 - \sum_{r=0}^{k-1} \frac{e^{-\lambda x} (\lambda x)^r}{r!} & \text{if } x > 0 \\ 0 & \text{otherwise.} \end{cases}$$

$$f(x) = \frac{d}{dx}F(x) = \begin{cases} \frac{e^{-\lambda x} (\lambda x)^{k-1}}{(k-1)!} & \text{if } x > 0 \\ 0 & \text{otherwise.} \end{cases}$$

# Example distributions

## 3. Uniform with parameters $a, b$

$$f(x) = \frac{d}{dx}F(x) = \begin{cases} \frac{1}{b-a} & \text{if } a \leq x \leq b \\ 0 & \text{otherwise.} \end{cases}$$

$$F(x) = P(X \leq x) = \begin{cases} 0 & \text{if } x < a \\ \frac{x-a}{b-a} & \text{if } a \leq x \leq b \\ 1 & \text{if } x > b. \end{cases}$$

## 4. Normal random variable with parameters $\mu, \sigma^2$

$$f(x) = \frac{d}{dx}F(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, \quad -\infty < x < \infty.$$

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(u)du.$$

## Class problem 2

2. The lifetime of a single-cell organism is exponentially distributed with parameter 0.1 (per hour). What is the probability that this organism will live for more than 20 hours? What is the probability that it will die within 5 hours?
  
3. The time for a message to reach `www.microsoft.com` from POSTECH is distributed according to an Erlang distribution with mean 300 ms and variance 30000 squared-ms. What is the probability that a message will reach within 600 ms? What is the probability it will take more than 900 ms?

► Colab notebook: CP2 §2–3 (exponential, Erlang)

# Expectation of a random variable

$$\begin{aligned} E[X] &= \int_{\Omega} X(\omega) dP(\omega) \\ &= \begin{cases} \sum_x x \cdot p(x) \\ \int_{\mathbb{R}} x dF(x) = \int_{\mathbb{R}} x \cdot f(x) dx. \end{cases} \quad \text{if } x \text{ is discrete} \end{aligned}$$

▷ Tail-sum, linearity, LOTUS: Companion, Proofs 3–5.



# Expectation of a random variable

- Example: Poisson random variable  $X$  with  $\lambda$ .

▷ Result  $E[X] = \lambda$ : Companion §1.6.

# Expectation of a random variable

- Example: Exponential random variable  $X$  with  $\lambda$ .

▷ One-line route via tail-sum: Companion, Proof 3 (checkpoint).

# Expectation of a function of a random variable

$$E[g(X)] = \begin{cases} \sum_x g(x)p_X(x) & \text{if } x \text{ is discrete} \\ \int_{\mathbb{R}} g(x)f_X(x)dx & \text{if } x \text{ is continuous.} \end{cases}$$

- Warning: in general

$$E[g(X)] \neq g(E[X]).$$

▷ LOTUS proof: Companion, Proof 5.

# Class problem 2

4. If the cdf of a continuous r.v. is given by

$$F(x) = \begin{cases} 0 & \text{if } x < 0, \\ ax + \frac{bx^3}{3} & \text{if } 0 \leq x \leq 1, \\ 1 & \text{if } x > 1. \end{cases}$$

If  $E[X] = 0.6$ , find  $a$  and  $b$ .

► Colab notebook: CP2 §4 (find  $a, b$ )

## Class problem 2

5. If  $X$  is a continuous r.v. denoting the size of a document at a web server with pdf

$$f(x) = \begin{cases} \frac{24}{x^4} & \text{if } x \geq 2, \\ 0 & \text{otherwise.} \end{cases}$$

Compute the variance of  $X$ .

► Colab notebook: CP2 §5 (variance, heavy tail)